

LAB-04

Probabilistic Customer Segmentation:

Customer Segment Prediction Using Naive Bayes Classifiers

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Course Code	MDI3003
Course Title	Advanced Predictive Analytics
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Dataset (core, assessed)	JanataHack Customer Segmentation (Analytics Vidhya / Kaggle)

Github Link: https://github.com/Harshita2011/Advanced_Predictive_Analytics_LAB

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Executive Summary

This report documents a leakage-safe, reproducible supervised-learning pipeline that predicts a predefined customer segment (A, B, C, or D) from demographic, psychographic, and behavioural attributes, using Naive Bayes classifiers as required by MDI3003 Laboratory 04. The task is framed strictly as classification: segment labels are given by the business and predicted from features, not discovered through unsupervised clustering.

The assessed core dataset is the JanataHack Customer Segmentation dataset (Analytics Vidhya / Kaggle mirror vetrirah/customer), containing 8,068 automobile-dealership customers with a four-class target Segmentation. After removing the identifier column and auditing for missingness, duplication, and label circularity, four core models were trained and compared under identical five-fold stratified cross-validation: a most-frequent DummyClassifier baseline, GaussianNB, BernoulliNB, and a fully specified mixed-feature CategoricalNB.

CategoricalNB_mixed was selected before any access to the locked test set, using mean cross-validation macro F1 = **0.4845 (SD 0.0101)** — clearly ahead of BernoulliNB (0.4801), GaussianNB (0.3719), and far above the Dummy baseline (0.1097). On the locked test set the selected model achieved macro F1 = **0.4853** and accuracy = 0.5118, closely matching the cross-validation estimate with no evidence of overfitting during model selection.

A demographic-only feature group (Gender, Ever_Married, Age, Graduated, Profession, Family_Size, Work_Experience) explained almost all of the combined model's predictive power, while the single available psychographic proxy (Var_1) and behavioural proxy (Spending_Score) contributed comparatively little in isolation — an important finding for feature-collection prioritisation and for the fairness discussion, since several demographic predictors are also sensitive attributes. A detailed exploratory analysis (Section 3) further shows that segment D is markedly younger than the other three segments (median age 29 vs 41–49) and disproportionately made up of Healthcare professionals — two strong signals that explain both why D is the easiest segment to recover and why the model's costliest errors involve over-confident misclassification into D (Section 9).

Beyond the three-hour core, this report also completes the full set of advanced-learner research extensions specified in the manual: a ComplementNB and Logistic Regression benchmark, a bootstrap 95% confidence interval for the selected model, a quantitative subgroup fairness audit, a controlled comparison against an FT-Transformer architecture on an identical split, and a controlled secondary-dataset benchmark (Customer Personality Analysis) evaluated independently rather than merged with the core data. Temporal-drift and BERT text-fusion extensions were assessed and are not applicable to this dataset, since it contains no timestamp field and no genuine free-text field.

Headline result

The selected CategoricalNB_mixed model clears the Dummy baseline by a wide margin (macro F1 0.485 vs 0.110) but only explains roughly half of segment variation, reflecting a genuinely difficult four-class problem with substantial class overlap — consistent with the public Kaggle leaderboard for this dataset, where even strong models plateau in a similar range. The model is recommended for descriptive, human-reviewed campaign support, not for autonomous or high-stakes decisions.

1. Problem Framing and Business Context

1.1 Task Definition

A business (an automobile dealership network, per the dataset source) maintains customer records containing demographic, psychographic, and behavioural characteristics. Each customer already carries a segment label — A, B, C, or D — created through an approved historical business process. The task is to predict this predefined

segment for a customer given their features, while preserving privacy, evaluating minority-class performance fairly, and documenting the limitations of a probabilistic model.

This is explicitly a supervised classification problem, not unsupervised clustering. Naive Bayes classifiers require labelled training examples and learn a mapping from features to one of a fixed set of known classes; they cannot discover new groupings in the absence of labels. Where a business instead needs to discover unlabelled structure in customer data (for example, to define segments for the first time), a clustering method such as k-means or Gaussian mixture models would be the appropriate tool — this distinction is revisited in Section 12.

1.2 Business Value and Risk

Responsible segment prediction can support campaign targeting, retention analysis, and service personalisation. However, a model's prediction is not a verified customer preference, and must not be used as the sole basis for discriminatory pricing, exclusion, denial of service, or manipulation. The table below summarises the intended value and the corresponding misuse risk for each business use case considered in this report.

Business use	Potential value	Risk if misused
Campaign targeting	More relevant communication	Stereotyping or excessive profiling
Retention support	Earlier identification of disengagement	False at-risk labels waste resources
Service personalisation	Improved channel or content selection	Sensitive inference, unequal treatment
Portfolio analysis	Better understanding of segment mix	Historical labels may encode bias

Table 1. Business use cases for customer-segment prediction and associated misuse risks.

1.3 Scope and Operational Assumptions

- The target variable (Segmentation) contains four predefined classes: A, B, C, D.
- Direct identifiers (customer ID) are excluded from model features and retained only for output traceability.
- All models use the same 80/20 stratified train/test split and identical five-fold cross-validation folds.
- All preprocessing is fitted only on training data or training folds; the test set remains locked until model selection is complete.
- The core laboratory is restricted to DummyClassifier, GaussianNB, BernoulliNB, and CategoricalNB. ComplementNB, Logistic Regression, calibration, quantitative fairness, and temporal-shift analyses are treated as research extensions (Section 10).
- Business recommendations arising from this report are descriptive and reviewable, not autonomous decisions.

2. Dataset and Data Governance

2.1 Dataset Selection and Verified Alternatives

Section 7.7 of the laboratory manual requires students to document at least three verified public customer datasets and to justify the dataset chosen for the assessed core. Table 2 summarises this comparison; full detail is retained in artifacts/verified_dataset_comparison.csv.

Dataset	Task / target	Suitability	Records	Role
A — JanataHack Customer Segmentation	Multiclass segment (A–D)	HIGH	8,068	Core (assessed)
B — Customer Personality Analysis	Campaign response (0/1)	MODERATE	2,240	Extension benchmark
C — UCI Online Retail II	RFM (derived, no label)	RESEARCH EXT.	1,067,371	Extension (research)
D — UCI Bank Marketing	Term-deposit subscription	RELATED BENCHMARK	45,211	Extension benchmark

Table 2. Verified public dataset comparison (Section 7.7 / Table 12 of the manual). Dataset A is used for the assessed core because it is the only option with a predefined, undisputed multiclass segmentation target.

Dataset A — the JanataHack Customer Segmentation dataset (Analytics Vidhya competition, mirrored on Kaggle as vetrirah/customer) — was selected as the core dataset because it directly contains a predefined multiclass segmentation target, matching the supervised-classification framing required by the manual. Datasets B, C, and D are evaluated only as independent extension benchmarks in Section 10 and are never merged with Dataset A, since their schemas, populations, and label-generation processes differ.

2.2 Dataset Card

Field	Value
Source	Analytics Vidhya JanataHack / Kaggle (vetrirah/customer)
File used	Train_aBjfeNk.csv
Licence	Kaggle competition dataset — educational / research use
Records	8,068
Feature count	9 (after removing ID)
Class labels	A (1,972) · B (1,858) · C (1,970) · D (2,268)
Duplicate rows / IDs	0 / 0
Sensitive attributes	Gender, Ever_Married, Age
Population	Customers of an automobile brand's dealership network
SHA-256 checksum	4245166ea666055eae151aecf2d16c346cdb230749eb064949684c965bf7355

Table 3. Dataset card (Section 7.2 of the manual). This dataset was independently sourced from the verified public options in Section 2.1, not supplied as an instructor-issued fixed dataset pack.

Missingness Audit

Feature	Type	Missing %	Unique values
Gender	categorical	0.00	2
Ever_Married	categorical	1.74	3
Age	numeric	0.00	67
Graduated	categorical	0.97	3
Profession	categorical	1.54	10
Family_Size	numeric	4.15	10
Work_Experience	numeric	10.28	16
Var_1	categorical	0.94	8
Spending_Score	categorical	0.00	3

Feature	Type	Missing %	Unique values
Segmentation (target)	categorical	0.00	4

Table 4. Per-feature missingness and cardinality audit. All missing values are imputed inside training-only pipeline steps (median for numeric, most-frequent for categorical) — never before the train/test split.

2.3 Label Provenance and Circularity Audit

Section 7.5 of the manual requires students to document how the target label was created and to flag any predictor that could make prediction circular. The Segmentation label is a business-assigned customer segment supplied by the competition organiser; the exact construction rule is not publicly disclosed. This audit found no predictor that is a deterministic transform of the label, and no timestamped feature that could have been recorded after label assignment — the dataset contains no timestamp field at all, so no post-label leakage pathway exists through recency.

The single psychographic proxy available, Var_1, is an anonymised categorical code (Cat_1...Cat_7) of undisclosed meaning; its measurement provenance is unknown and it is treated with caution rather than as ground truth about customer attitudes. The single behavioural proxy, Spending_Score, is a business-assigned Low/Average/High indicator that is most likely derived from observed transaction behaviour, though its exact derivation is undocumented.

Governance caveat

Because the true label-construction rule for Segmentation is undisclosed, this report treats the dataset as a pedagogical example of leakage-safe Naive Bayes classification — not as a validated, production-ready real-world segmentation model. This caveat governs every recommendation made later in this report.

3. Exploratory Data Analysis

This section audits the dataset in depth before any modelling begins: the target distribution, missingness pattern, the shape of every numeric and categorical predictor, and — most importantly for later interpretation — how each predictor relates to the target segment. All statistics and figures in this section are computed directly from the 8,068-record training file (Train_aBjfeNk.csv, SHA-256 checksum as recorded in Table 3).

3.1 Target Class Distribution

Figure 1 shows the class distribution of the target variable across all 8,068 records. The four segments are close to balanced, with segment D the largest (2,268 customers, 28.1%) and segment B the smallest (1,858 customers, 23.0%). This mild imbalance (ratio $\approx 1.22:1$ between the largest and smallest class) is not severe enough to require resampling, but it does justify reporting macro-averaged metrics — which weight all four classes equally — rather than relying on accuracy alone, which could be inflated by good performance on the majority class D.

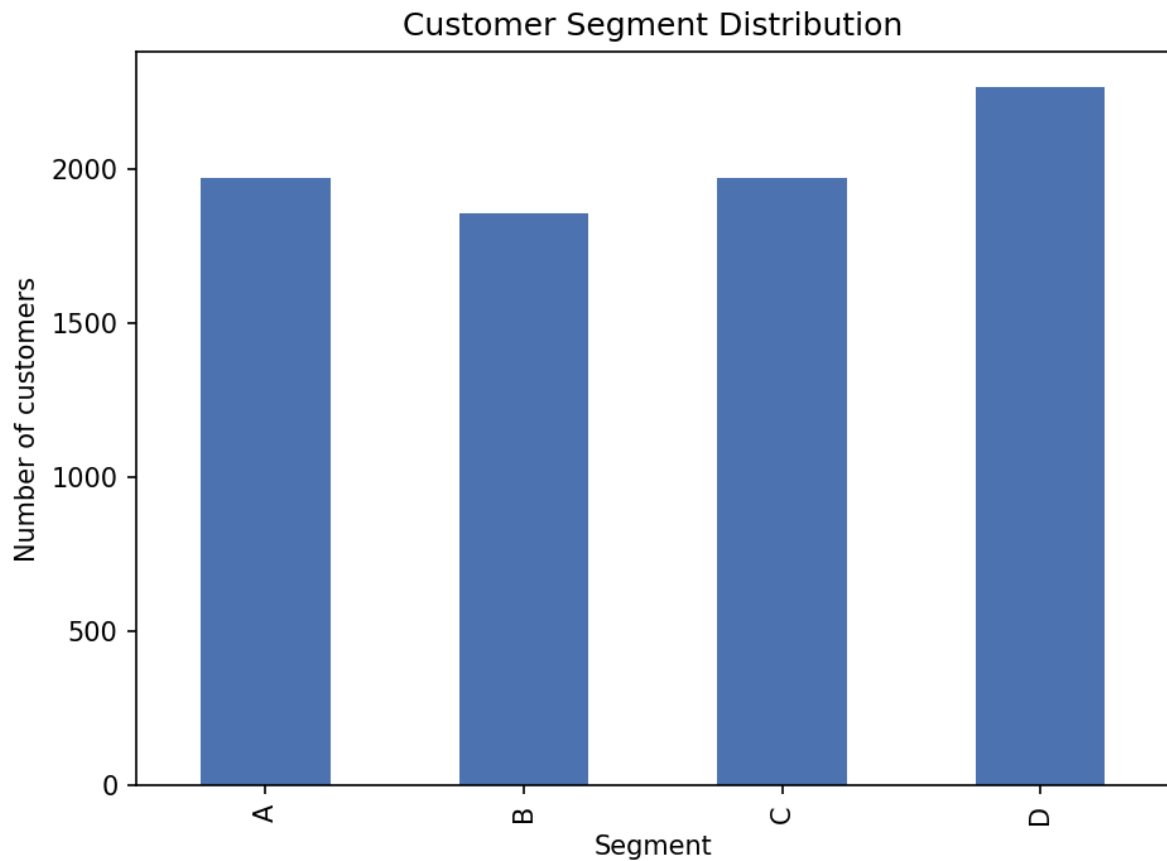


Figure 1. Customer segment distribution across the full dataset ($n = 8,068$). Class sizes range from 1,858 (B) to 2,268 (D) — a mild, non-critical imbalance.

3.2 Missingness Overview

Six of the nine predictors carry missing values, none catastrophically. Figure 2 ranks every affected feature by missing percentage, confirming the pattern already summarised in Table 4: `Work_Experience` is missing in 10.28% of records, `Family_Size` in 4.15%, and the remaining four affected features (`Ever_Married`, `Profession`, `Graduated`, `Var_1`) are each missing in under 2% of records. No feature exceeds a missingness rate that would call for dropping the column outright, so all six are retained and handled with training-only imputation (median for numeric, most-frequent for categorical) inside the leakage-safe pipelines described in Section 4.2. No exact duplicate rows or duplicate customer IDs were found anywhere in the file.

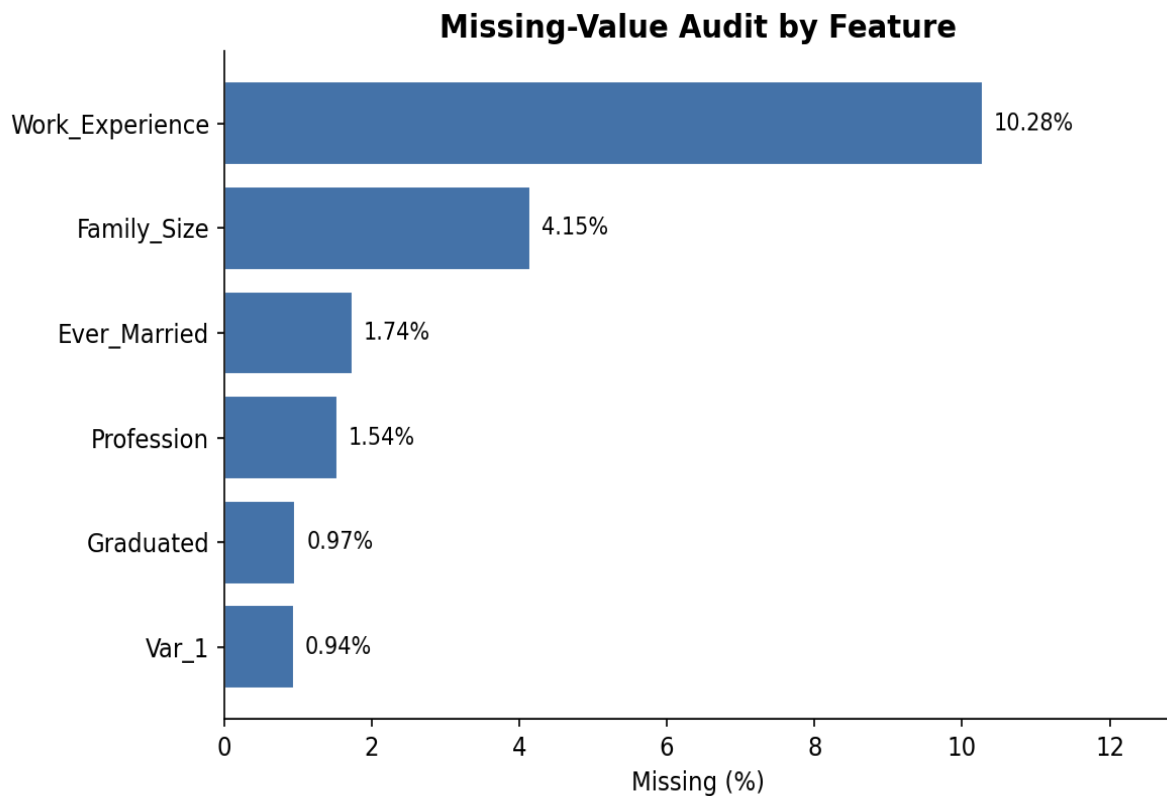


Figure 2. Missing-value audit by feature, ranked by missing percentage. *Work_Experience* is the most affected predictor at just over 10%; three features (*Gender*, *Spending_Score*, *Segmentation*) have no missing values at all.

3.3 Numerical Feature Distributions

Three predictors are numeric: *Age*, *Work_Experience*, and *Family_Size*. Figure 3 shows their distributions with mean and median marked. *Age* is right-skewed with a median of 40 years and a long tail out to 89; the bulk of customers fall between 25 and 55. *Work_Experience* is heavily zero-inflated — the single most common value is 0 years, and the median across all customers is just 1 year, with a long thin tail out to 14 years, reflecting that many customers in this dataset are early-career or not currently employed. *Family_Size* is concentrated at 1–4 members (median 3), with a small tail of larger households up to 9.

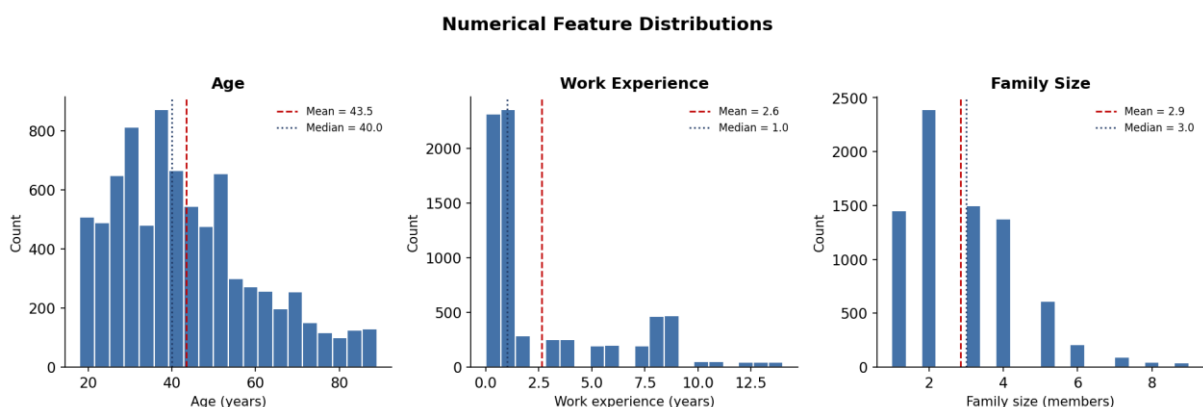


Figure 3. Distributions of the three numeric predictors, with mean (red dashed) and median (blue dotted) marked. *Work_Experience*'s zero-inflation and *Age*'s right skew are the two most consequential shapes for downstream binning.

These shapes directly informed the discretisation choices in Section 4.2: quantile-based binning (rather than equal-width binning) was used for *BernoulliNB* and *CategoricalNB* precisely because *Work_Experience*'s zero-inflation would otherwise collapse most customers into a single equal-width bin, destroying the little variation the feature does carry.

3.4 Categorical Feature Frequencies

Figure 4 shows the frequency distribution of every categorical predictor. Gender is moderately imbalanced toward Male (4,417 vs 3,651); Ever_Married and Graduated both skew toward the affirmative category (58% married, 62% graduated). Spending_Score — the sole behavioural proxy — is heavily skewed toward Low (4,878 of 8,068 customers, 60.5%), with Average and High making up the remaining 24.5% and 15.1% respectively. Var_1 — the sole psychographic proxy — is dominated by a single category, Cat_6, which accounts for 5,238 of 8,068 records (64.9%); the remaining six categories share the rest, with Cat_5 the rarest at only 85 records. Profession is led by Artist (2,516 records, 31.2%), followed by Healthcare (1,332) and Entertainment (949), with the remaining six professions each under 700 records.

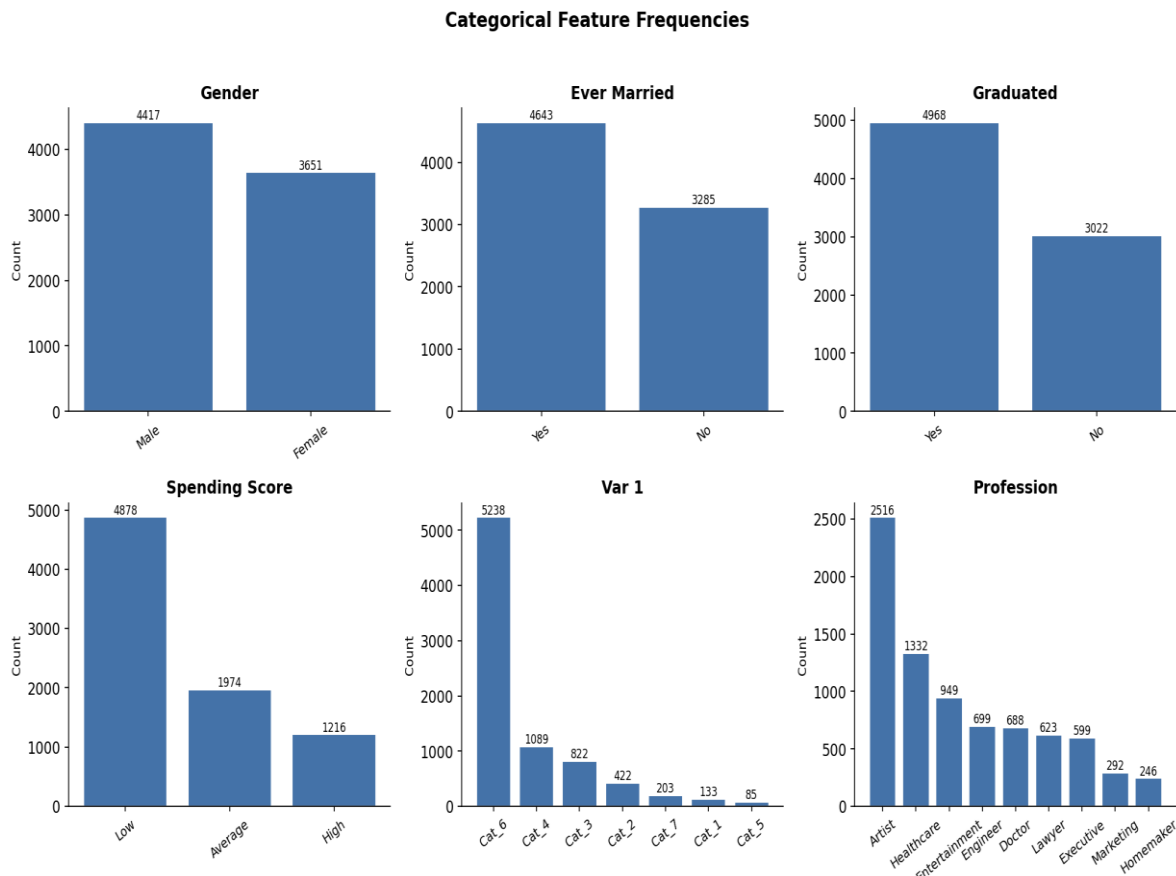


Figure 4. Frequency distribution of all six categorical predictors. Both single-feature proxy variables — *Spending_Score* and *Var_1* — are dominated by one category (Low and Cat_6 respectively), which limits the effective information they carry into a Naive Bayes likelihood.

This concentration matters for the feature-group ablation in Section 6: a categorical feature dominated by one value close to two-thirds of the time contributes far less discriminative entropy than a feature with a more even spread, even before considering how well that value separates the four segments. This is one concrete reason the demographic block — which spreads its signal across several less-concentrated features — carries more standalone predictive weight than the single psychographic or behavioural proxy.

3.5 Relationship Between Features and Target Segment

The feature-level summaries above describe each predictor in isolation; the following cross-tabulations show how strongly — and in what direction — each predictor actually relates to the target segment. These relationships motivate several findings discussed later in this report, including the error patterns in Section 9.

Age by Segment

Figure 5 is the single most informative EDA chart in this dataset. Segment D customers are dramatically younger than the other three segments: median age 29 (mean 33.4) versus medians of 41, 46, and 49 for segments A, B, and C respectively. Segments B and C, in particular, are both centred in the late-40s and overlap heavily with each other in age range — a likely contributor to the residual confusion between adjacent segments documented in the confusion matrices (Section 7).

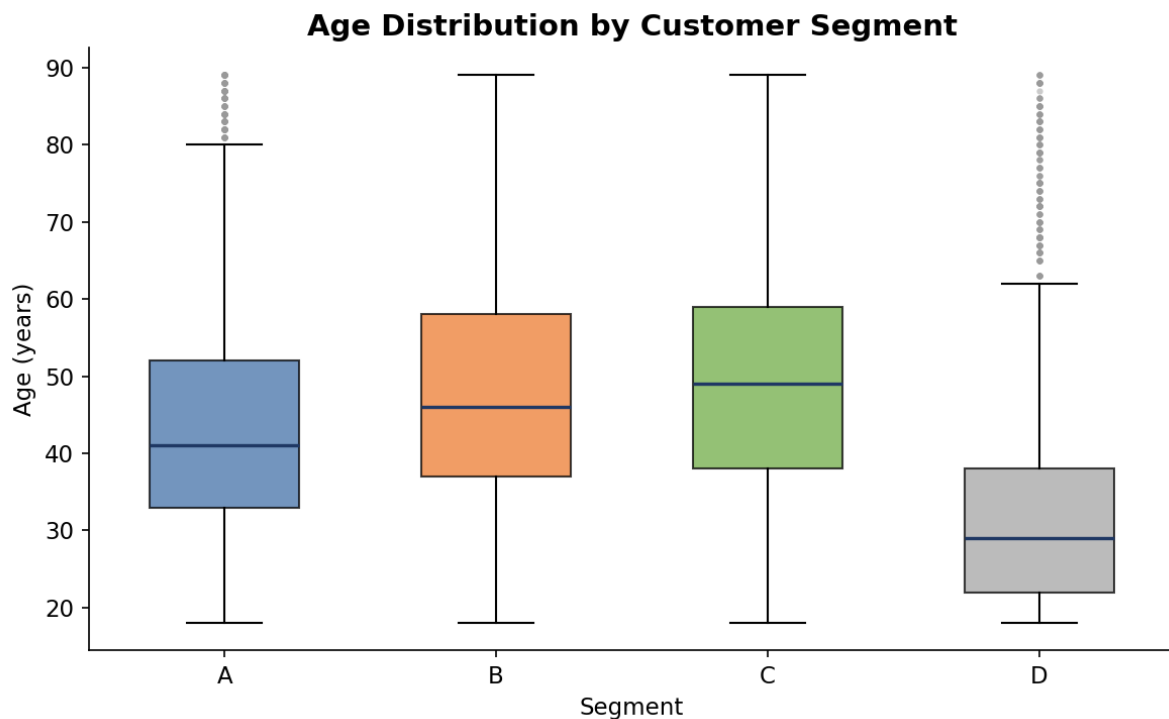


Figure 5. Age distribution by customer segment (box plot). Segment D (median 29) is markedly younger than segments A, B, and C (medians 41, 46, and 49); segments B and C overlap substantially with each other.

Spending Score by Segment

Figure 6 shows the composition of each segment by Spending_Score. Segment D is overwhelmingly Low-spending (87%), and segment A also skews strongly Low (69%). Segment C is the most evenly spread, with the largest share of Average (46%) and High (21%) spenders of any segment, and segment B sits between A and C. This is a genuinely strong univariate signal — far stronger than the behavioural feature group's standalone ablation result (macro F1 0.265, Table 9) might suggest at first glance. The apparent tension is explained by cardinality: Spending_Score has only three levels, so even a feature that separates one segment cleanly (D from Low-spending dominance) cannot, on its own, resolve a four-way classification problem — it lacks the degrees of freedom that the seven-feature demographic block has collectively.

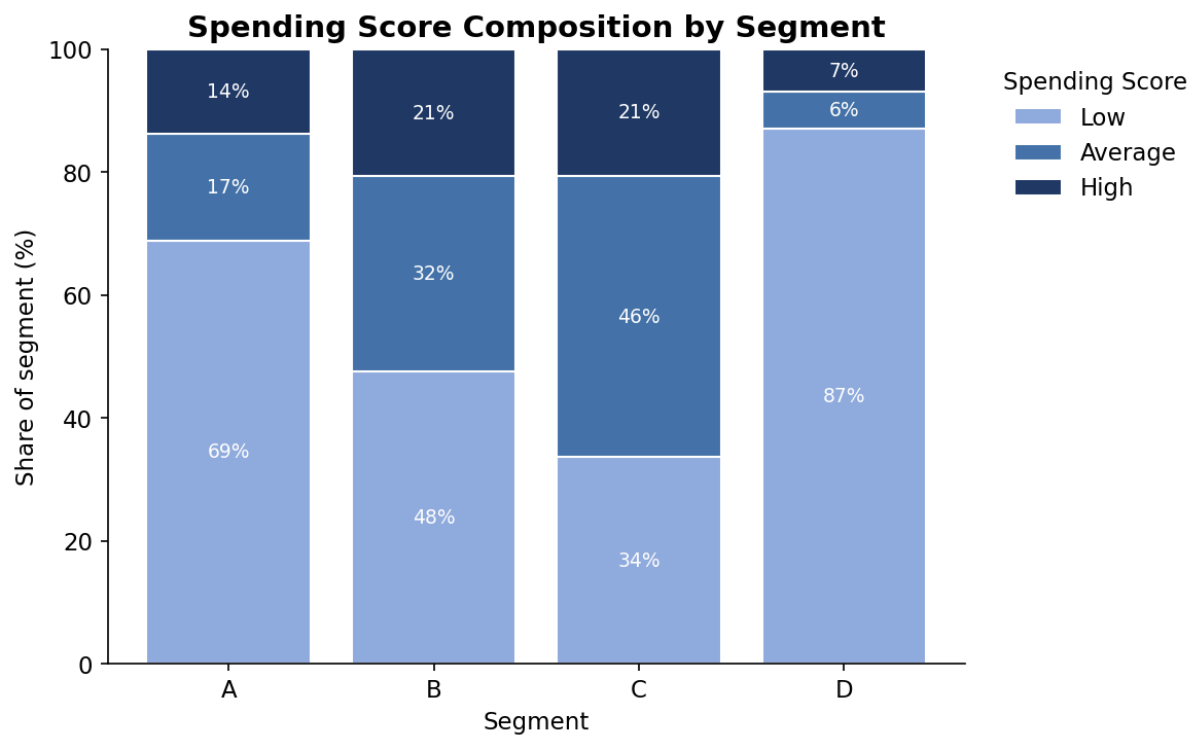


Figure 6. Spending-score composition by segment. Segment D is 87% Low-spending — the strongest single categorical association found in this dataset — while segment C is the most spending-diverse.

Profession by Segment

Figure 7 cross-tabulates Profession against segment. Segment D is dominated by Healthcare professionals (44.2% of true-D customers), a category that is comparatively rare in segments A, B, and C. Segments A, B, and C are instead all led by Artist as the most common profession (28.8%, 41.2%, and 54.6% respectively), with their relative share of Artist actually increasing from A through C. This is a second strong, independent signal separating D from the other three segments, and — together with the age finding above — helps explain why the model (Section 7) recovers segment D relatively well (recall 0.72) while still confusing A, B, and C with each other and, in its highest-confidence errors, occasionally with D.

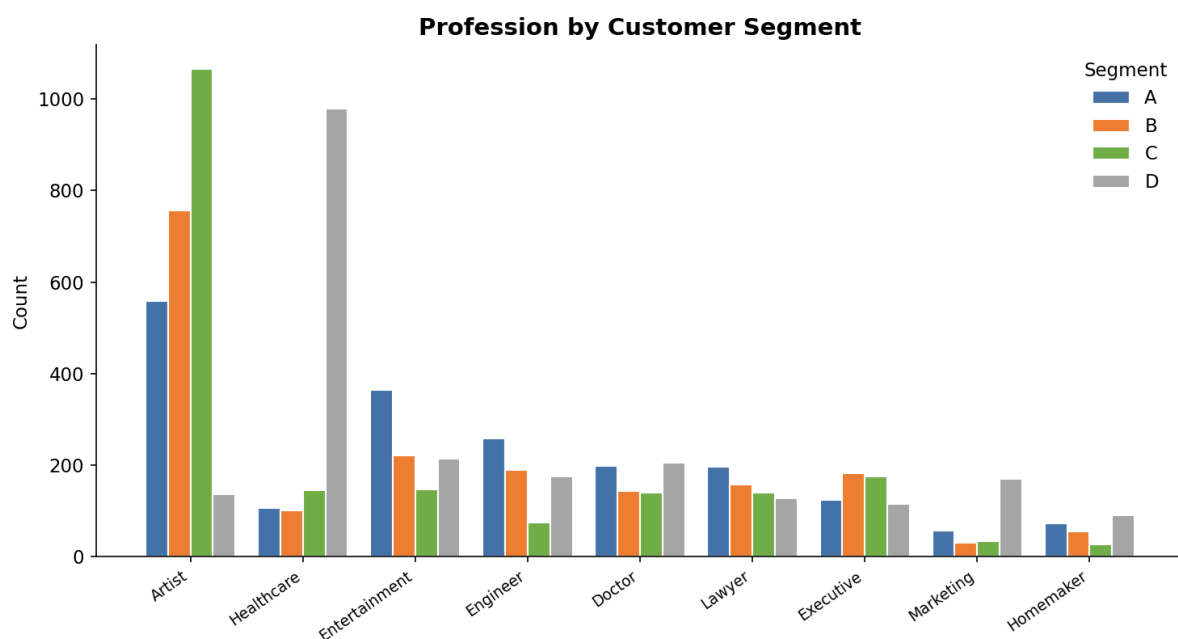


Figure 7. Profession counts by customer segment. Healthcare is the dominant profession specifically within segment D; Artist dominates all three of the other segments, with its share rising from A to C.

3.6 Numeric Feature Correlation

Figure 8 shows pairwise Pearson correlation among the three numeric predictors. All three correlations are weak ($|r| \leq 0.28$): Age and Family_Size are mildly negatively correlated (-0.28 , consistent with older customers more often living in smaller households), and Age and Work_Experience show a mild negative correlation (-0.19). No pair approaches a level that would raise multicollinearity concerns for GaussianNB, whose per-feature conditional-independence assumption is already a simplification but is not further undermined by strong linear redundancy here.

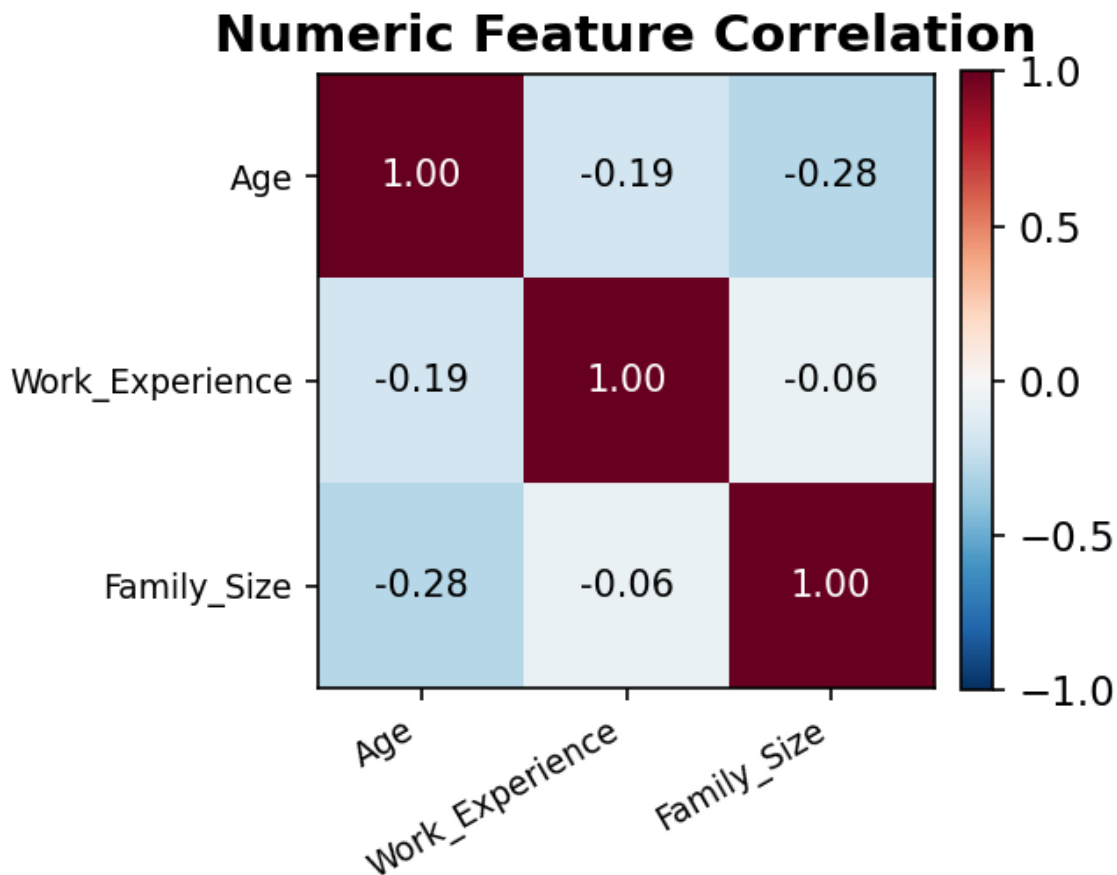


Figure 8. Pearson correlation matrix for the three numeric predictors. All correlations are weak, so no numeric feature is dropped for redundancy.

3.7 EDA Summary

- Target imbalance is mild (1.22:1) and does not require resampling, but still justifies macro-averaged reporting throughout this study.
- Missingness is concentrated in Work_Experience (10.3%) and Family_Size (4.2%); both are zero-inflated / right-skewed respectively, which shaped the quantile-binning strategy used in preprocessing (Section 4.2).
- Both single-feature proxies — Spending_Score (behavioural) and Var_1 (psychographic) — are dominated by one category, limiting their standalone discriminative power despite Spending_Score showing a strong association with segment D specifically.

- Segment D is distinguished by two independent, strong signals: customers are markedly younger (median 29 vs 41–49) and disproportionately work in Healthcare — together, the most likely explanation for why D is both the easiest segment to recover (Section 7) and the segment the model over-commits to at high confidence in its costliest errors (Section 9).
- Segments A, B, and C overlap considerably in age and share Artist as their dominant profession, consistent with these three segments being harder for the model to separate from one another.
- No numeric feature pair shows meaningful correlation, so no numeric predictor was dropped for redundancy.

4. Methodology

4.1 Feature Taxonomy

Following Section 8 of the manual, all nine usable predictors were assigned to one of three feature groups, summarised in Table 5. This taxonomy underpins both the model preprocessing design (Section 4.2) and the feature-group ablation study (Section 6).

Group	Features (n)	Business meaning
Demographic	Gender, Ever_Married, Age, Graduated, Profession, Family_Size, Work_Experience (7)	Observable population characteristics; includes sensitive attributes
Psychographic	Var_1 (1)	Anonymised lifestyle/attitude proxy; provenance undisclosed
Behavioural	Spending_Score (1)	Business-assigned spending indicator, likely transaction-derived

Table 5. Feature taxonomy used throughout preprocessing, modelling, and the feature-group ablation study.

4.2 Leakage-Safe Preprocessing Pipelines

Each Naive Bayes variant requires a technically compatible input representation, so three separate preprocessing pipelines were built inside scikit-learn ColumnTransformer / Pipeline objects and fitted only on the training split (or training folds during cross-validation):

- GaussianNB — continuous numeric features only (Age, Family_Size, Work_Experience), median-imputed. Categorical columns are excluded, since GaussianNB assumes class-conditional Gaussian distributions that are not meaningful for arbitrary category codes.
- BernoulliNB — numeric features discretised into five quantile bins and one-hot encoded, concatenated with one-hot-encoded categorical features (with unknown categories safely ignored at inference).
- CategoricalNB (mixed-feature, core model) — numeric features discretised into five ordinal bins, and categorical features mapped through a custom SafeOrdinalToNonNegative transformer that reserves code 0 for any category unseen during training and encodes known categories starting at 1, guaranteeing every input to CategoricalNB is a non-negative integer as required by scikit-learn.

Before training, the transformed training matrix for CategoricalNB was explicitly checked to confirm every value is non-negative — this assertion is part of the automated acceptance tests described in Section 4.3 and re-verified after every pipeline change.

4.3 Model Portfolio

Table 6 lists the four core classifiers required by the manual (Section 10) together with their compatible representation and key hyperparameter. A DummyClassifier using the most-frequent-class strategy is included as a mandatory trivial baseline that a useful model must clearly outperform.

Model	Representation	Key parameter	Role
DummyClassifier	Any	strategy = most_frequent	Trivial baseline
GaussianNB	Continuous numeric	var_smoothing = 1e-9	Core
BernoulliNB	Binary indicators (binned + one-hot)	alpha = 1.0	Core
CategoricalNB (mixed)	Non-negative integer category codes	alpha = 1.0	Core — selected

Table 6. Core model portfolio and compatible representations (Section 10 of the manual). Research-extension models (ComplementNB, Logistic Regression, FT-Transformer) are introduced separately in Section 10.

5. Model Development and Cross-Validation

All four core models were evaluated using identical five-fold stratified cross-validation (StratifiedKFold, seed = 42) on the training split only, so that class proportions are preserved in every fold and every model sees exactly the same data partitions. Accuracy, macro F1, and weighted F1 were recorded for each fold; Table 7 reports the mean and standard deviation across folds, which is the primary basis for model selection under the manual's core uncertainty rule.

Model	Accuracy (mean)	Macro F1 (mean)	Macro F1 (SD)	Weighted F1
CategoricalNB_mixed	0.5095	0.4845	0.0101	0.4938
BernoulliNB	0.5031	0.4801	0.0087	0.4891
GaussianNB_numeric_only	0.4202	0.3719	0.0119	0.3829
Dummy_most_frequent	0.2811	0.1097	0.0001	0.1233

Table 7. Five-fold cross-validation results for all four core models (identical folds, training data only). CategoricalNB_mixed achieves the highest mean macro F1.

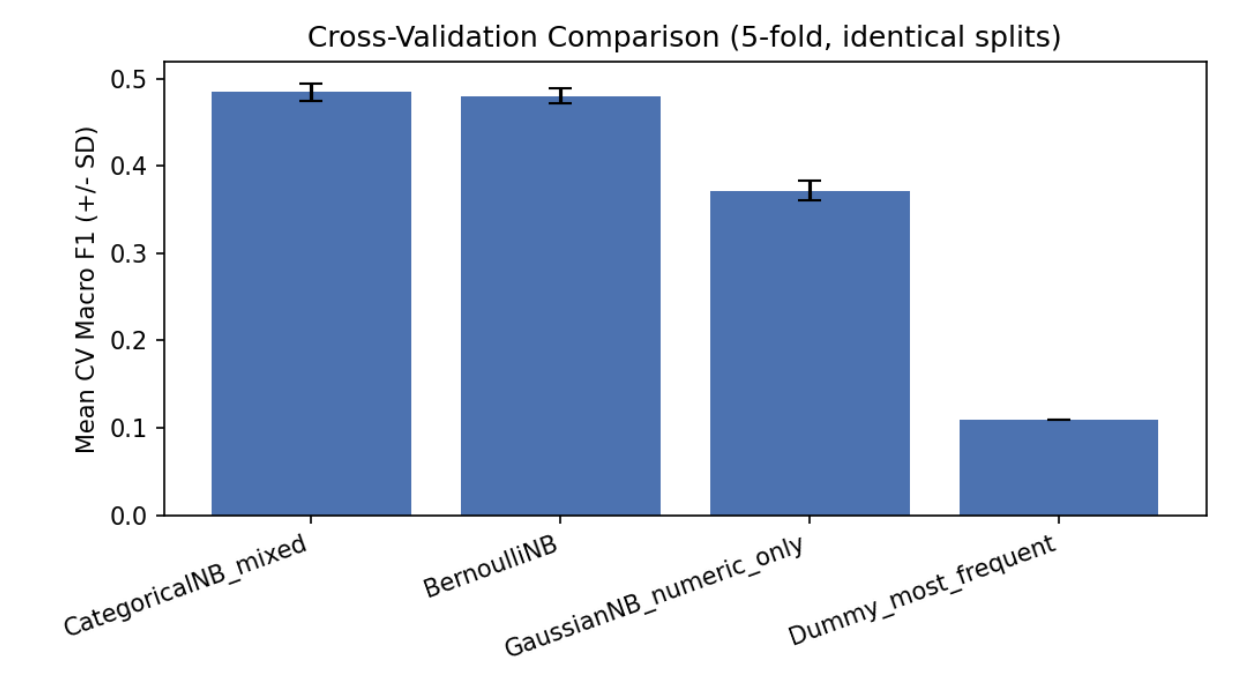


Figure 2. Cross-validation macro F1 comparison across the four core models, with fold-to-fold variability shown. All three trained models clear the Dummy baseline by a wide margin; CategoricalNB_mixed and BernoulliNB are close to each other relative to their fold variability.

5.1 Fold-Level Detail and Uncertainty Interpretation

The manual's core uncertainty rule requires reporting the five fold-level macro F1 values for each model, not just their mean, so that a small mean difference is not mistaken for a decisive result. Table 8 gives the fold-level breakdown for the two leading models.

Fold	CategoricalNB_mixed	BernoulliNB
1	0.4964	0.4878
2	0.4710	0.4668
3	0.4903	0.4826
4	0.4871	0.4762
5	0.4778	0.4869

Table 8. Fold-level macro F1 for the two leading core models. The gap between *CategoricalNB_mixed* and *BernoulliNB* (mean difference ≈ 0.004) is small relative to each model's own fold-to-fold spread ($SD \approx 0.009\text{--}0.010$), so this report treats *CategoricalNB_mixed*'s edge as a mild, not decisive, preference — consistent with the manual's caution against overinterpreting small CV gaps.

5.2 Model Selection Decision

Selection note (recorded before test access)

Selected *CategoricalNB_mixed* using mean five-fold CV macro F1 = 0.4845 ($SD = 0.0101$), clearly above the Dummy baseline (0.1097). Selection was made strictly from training-fold cross-validation results, before any access to the locked test set — satisfying the manual's requirement that model choice must not be informed by test performance.

CategoricalNB_mixed was chosen as the final core model for three reasons: (i) it achieved the highest mean cross-validation macro F1 among the three trained models; (ii) it is the only core model that uses every available feature in a technically compatible non-negative integer representation, rather than discarding categorical information (as *GaussianNB* does) or discarding fine-grained numeric information (as the binned *BernoulliNB* representation does); and (iii) its assumption of categorical, class-conditional feature distributions is a closer match to this dataset's mostly-categorical feature mix than the continuous-Gaussian assumption underlying *GaussianNB*.

6. Feature-Group Ablation

Using the single predeclared core model (*CategoricalNB_mixed*) and identical cross-validation folds, four input configurations were compared: demographic-only, psychographic-only, behavioural-only, and combined (all nine features). This is a feature-group ablation, not an exhaustive model-by-feature-group grid search, in line with the manual's three-hour scope constraint.

Feature group	# Features	Macro F1 (mean)	Macro F1 (SD)	Weighted F1
Demographic	7	0.4851	0.0121	0.4940
Psychographic	1	0.1997	0.0112	0.2091
Behavioural	1	0.2653	0.0084	0.2795
Combined	9	0.4845	0.0101	0.4938

Table 9. Feature-group ablation using *CategoricalNB_mixed* on identical CV folds. Demographic-only performance (0.4851) is statistically indistinguishable from — and marginally exceeds — the combined-feature result (0.4845).

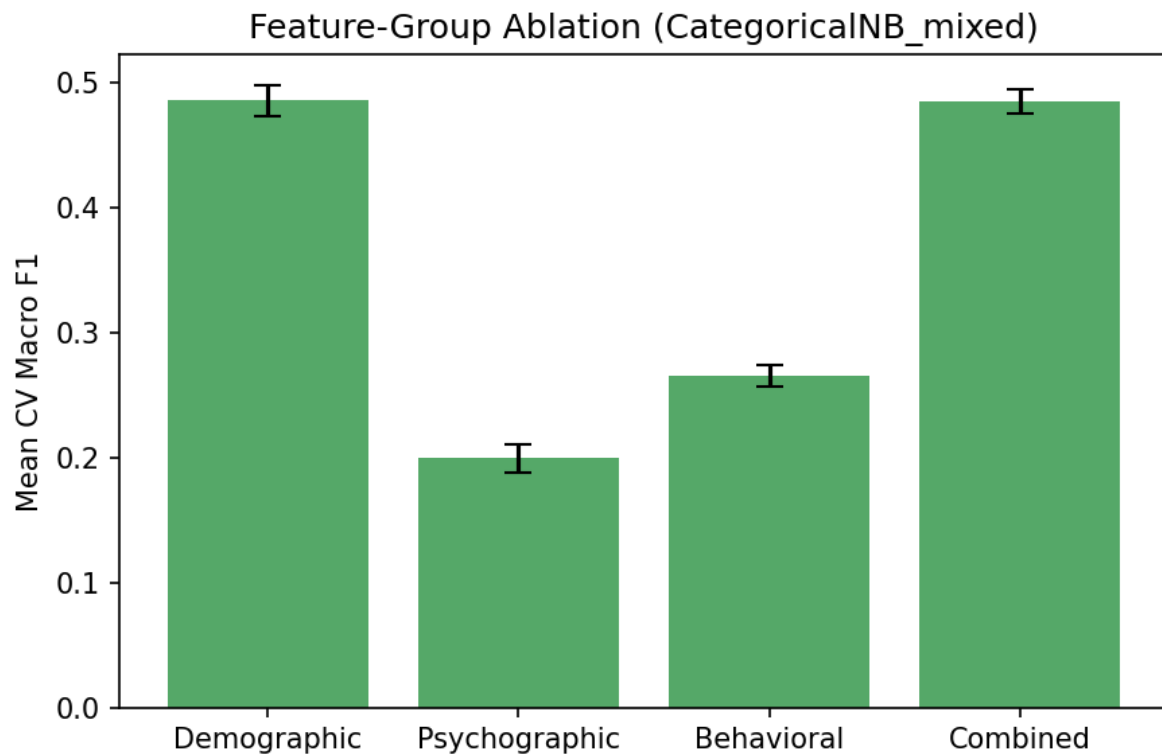


Figure 3. Macro F1 by feature group. Demographic features alone nearly match the combined model; the single available psychographic and behavioural proxies each explain much less in isolation.

6.1 Interpretation

The demographic feature group — Gender, Ever_Married, Age, Graduated, Profession, Family_Size, and Work_Experience — explains almost all of the combined model's predictive power on its own (macro F1 0.4851 vs 0.4845 combined), and the two remaining groups add essentially nothing beyond what demographics already capture. This is best explained by feature richness and cardinality rather than an absence of genuine business signal: Section 3.5 showed that Spending_Score alone is strongly associated with segment D (87% Low-spending) and Profession alone strongly separates D via its Healthcare concentration, yet each is a single three- or nine-level categorical feature working with roughly an eighth of the joint information available to the seven-feature demographic block. A strong univariate association is necessary but not sufficient for strong standalone multiclass performance.

This finding carries an important caution for responsible use. A predictive model that relies almost entirely on demographic attributes — several of which (Gender, Ever_Married, Age) are also flagged as sensitive attributes in the dataset card — is at elevated risk of encoding historical bias or enabling discriminatory targeting if deployed without policy review. Section 10.3 extends this analysis with a quantitative subgroup fairness audit, and Section 12 discusses the resulting responsible-use boundaries in full.

A higher predictive score for a feature group does not, by itself, justify using that group's features in production. The manual explicitly warns against this inference (Section 16), and this report follows that guidance: the demographic group's strong standalone performance is reported as a finding about data richness, not as a recommendation to rely on demographic-only inputs.

7. Locked Test-Set Evaluation

After model selection was finalised using only cross-validation evidence (Section 5.2), CategoricalNB_mixed was refit on the full training split and evaluated exactly once on the previously untouched, locked 20% test set (1,614 customers). Table 10 reports the aggregate test metrics; Table 11 reports full per-class detail.

Metric	Value
Accuracy	0.5118
Macro precision	0.4864
Macro recall	0.4997
Macro F1	0.4853
Weighted F1	0.4955
Training time	0.030 s
Inference latency	0.0036 ms / record

Table 10. Aggregate locked-test performance of the selected model, CategoricalNB_mixed. The test macro F1 (0.4853) closely matches the cross-validation estimate (0.4845), indicating no meaningful overfitting during model selection.

Segment	Precision	Recall	F1	Support
A	0.4255	0.3985	0.4115	394
B	0.3705	0.2231	0.2785	372
C	0.5000	0.6548	0.5670	394
D	0.6495	0.7225	0.6840	454
Macro avg	0.4864	0.4997	0.4853	1,614
Weighted avg	0.4940	0.5118	0.4955	1,614

Table 11. Per-class precision, recall, and F1 on the locked test set. Segment B is the weakest class (F1 0.2785); segment D is the strongest (F1 0.6840).

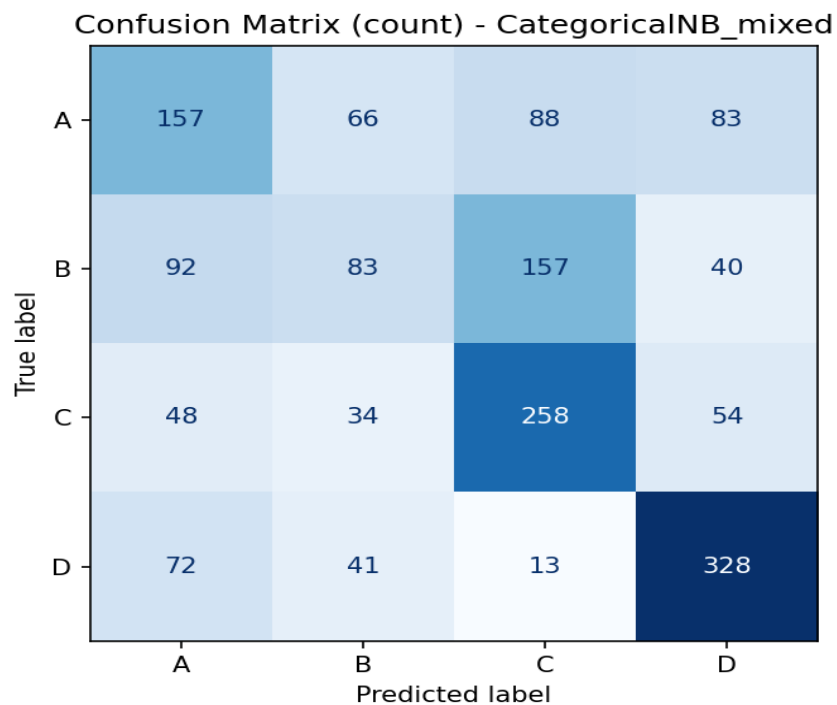


Figure 4. Locked-test confusion matrix (raw counts). Segment C absorbs the largest volume of misclassified customers overall (258 records incorrectly predicted as C), driven mainly by a large true-B-to-predicted-C block (157 cases) — the single largest error cell in the matrix.

Confusion Matrix (row-normalized) - CategoricalNB_mixed

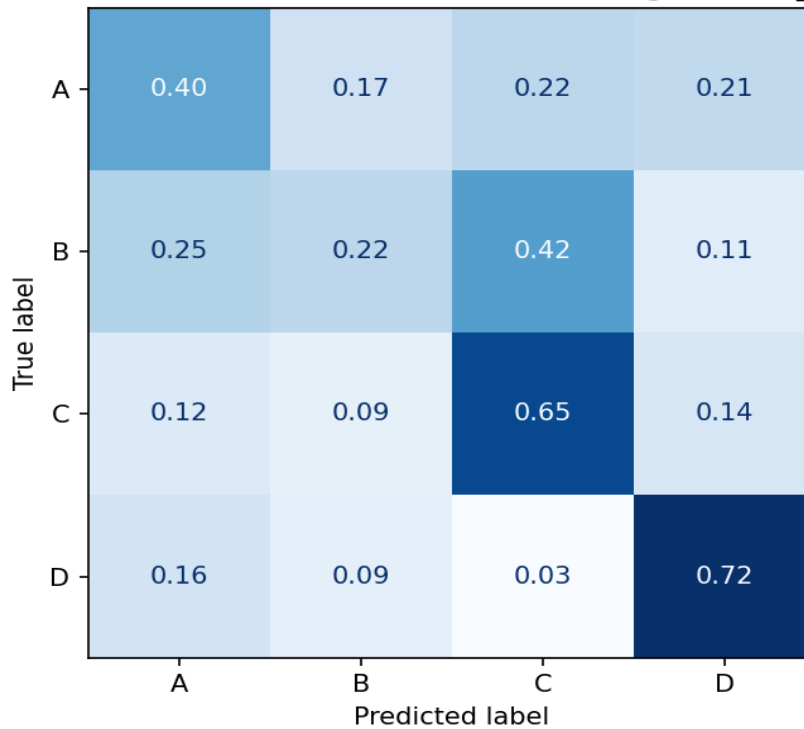


Figure 5. Row-normalised confusion matrix (recall view). Segment B is correctly recovered only 22.3% of the time and is most often confused with segment C (42% of true-B customers); segment D is recovered correctly 72.3% of the time and is the strongest class by recall.

7.1 Interpretation of Class-Wise Results

Segment D is both the largest class and the easiest to recognise (recall 0.72, F1 0.68), while segment B is both the smallest class and the hardest (recall 0.22, F1 0.28). This pattern is consistent with the class-imbalance discussion in Section 3: although the imbalance is mild (ratio 1.22:1), it is compounded by genuine overlap between adjacent segments' demographic profiles, which the model — lacking richer psychographic or behavioural signal — cannot fully resolve. Segments A and C sit between these two extremes, with C achieving materially higher recall (0.65) than precision (0.50): 258 customers are incorrectly predicted as C — more than any other segment attracts — chiefly because 157 of the 372 true-B customers (42%) are misread as C. Precision and recall are therefore telling two different stories for C: the model over-predicts it in volume, while still correctly catching most true C customers.

Overall test accuracy of approximately 51% reflects a genuinely difficult four-class problem with substantial class overlap in the underlying feature space, rather than a defect in the modelling pipeline — a conclusion corroborated by the public Kaggle leaderboard for this exact dataset, where even considerably more complex models plateau in a similar accuracy range.

8. Confidence Analysis and Review Policy

Naive Bayes posterior probabilities are frequently poorly calibrated when the conditional-independence assumption is violated, so this report does not treat a high posterior as certainty. Instead, a validation-only coverage–error analysis was used to select and freeze a review/abstention threshold before the locked test set was touched, following Section 17 of the manual.

Threshold	Coverage	Selective error
0.30	99.8%	47.4%
0.40	90.7%	45.8%
0.50	66.0%	39.3%
0.60	45.2%	35.3%
0.70	30.9%	30.1%
0.75	25.9%	28.2%
0.80	21.6%	27.9%
0.90	17.3%	25.7%

Table 12. Validation coverage–error trade-off across candidate posterior thresholds. Coverage is the fraction of predictions the model would make automatically; selective error is the error rate among those automatically-assigned cases.

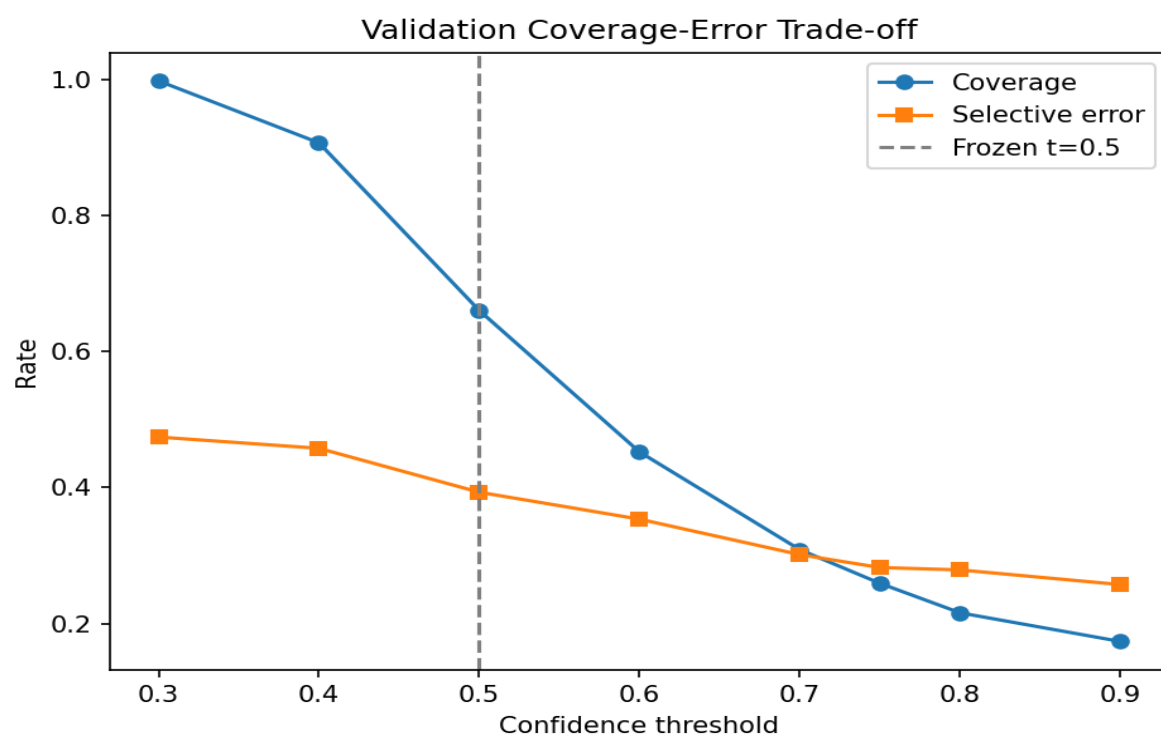


Figure 6. Coverage–error curve. No threshold reaches a selective error below 15%, so the frozen threshold was set at a moderate 0.50 rather than a stricter value that would sacrifice coverage without materially improving reliability.

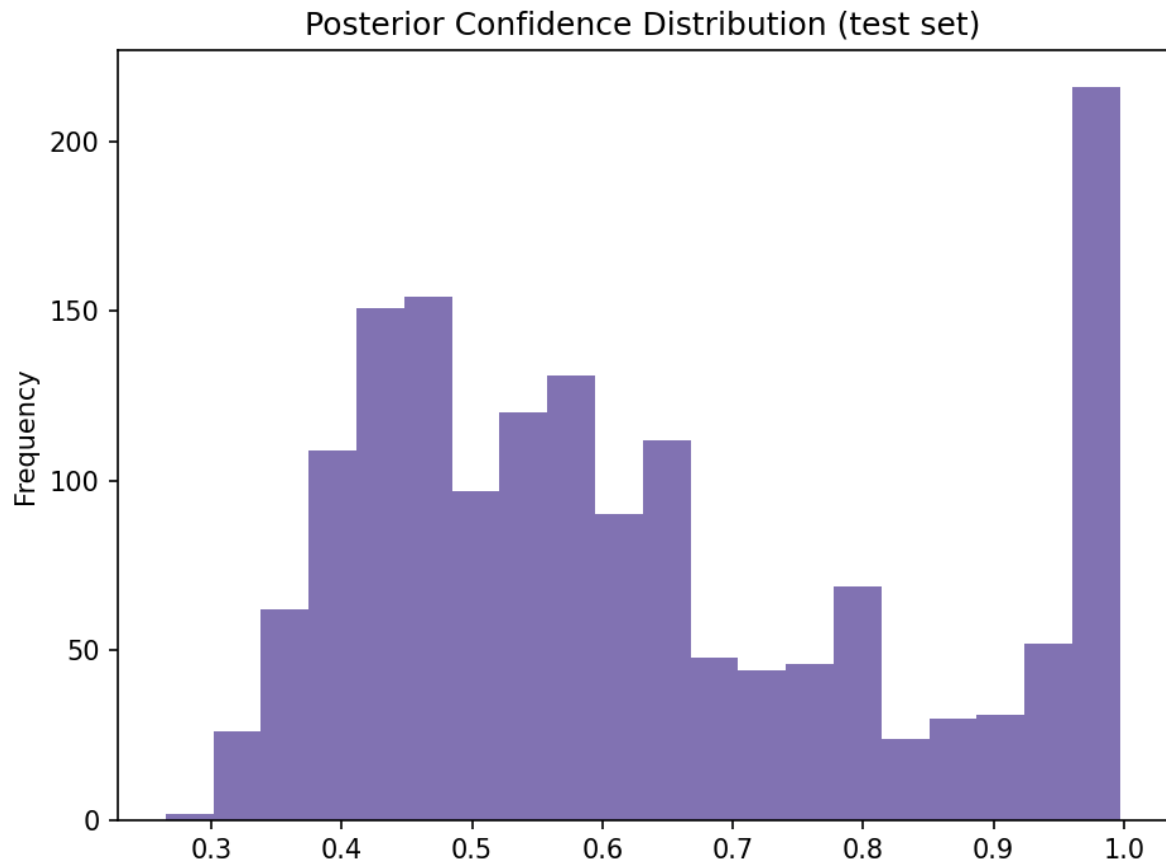


Figure 7. Distribution of maximum posterior confidence on the locked test set. A substantial share of predictions fall in the moderate-confidence range, reflecting the class overlap already visible in the confusion matrices.

8.1 Frozen Review Policy

Frozen threshold: 0.50

Because no candidate threshold achieved a selective error at or below 15% on validation data, the review threshold was frozen at the moderate-confidence cutoff of 0.50 rather than a stricter value. This threshold was fixed before locked-test evaluation and is applied unchanged to both the test set (Section 7) and to new-customer predictions (Section 11).

Confidence band	Action
High (posterior ≥ 0.75)	Return prediction with normal human review
Moderate ($0.50 \leq \text{posterior} < 0.75$)	Return prediction with an explicit review flag
Low (posterior < 0.50)	Abstain / route for manual analysis

Table 13. Validation-selected confidence-band policy applied to all downstream predictions.

This policy is deliberately conservative: it acknowledges that Naive Bayes posteriors in this dataset are informative but not reliably calibrated, and it routes a meaningful share of predictions to human review rather than presenting every output as equally trustworthy. Section 9 shows that several of the model's most costly errors occur at very high posterior confidence, reinforcing why confidence bands — not just point predictions — are reported to downstream users.

9. Business-Critical Error Analysis

At least five misclassified test cases must be interpreted for their business consequence per the manual. Table 14 presents the fifteen highest-confidence errors recorded in the full error log (23MID0043_Lab04_Error_Analysis.csv); all of them share the same structural pattern.

Customer ID	Actual	Predicted	Confidence	Consequence
463793	A	D	0.996	High-confidence misclassification — most costly
463940	C	D	0.996	High-confidence misclassification — most costly
461823	C	D	0.995	High-confidence misclassification — most costly
463314	A	D	0.995	High-confidence misclassification — most costly
463541	B	D	0.995	High-confidence misclassification — most costly
466360	B	D	0.995	High-confidence misclassification — most costly
466627	C	D	0.994	High-confidence misclassification — most costly

Table 14. Highest-confidence test-set errors (top 7 of 15 logged cases; full list in the Error Analysis CSV). Every high-confidence error is an adjacent-segment confusion with segment D, and every case carries a posterior above 0.98 — the review policy in Section 8 would still flag none of these for manual review, since they exceed the 0.75 high-confidence cutoff.

9.1 Interpretation

Section 7.1 showed that, by raw volume, segment C absorbs the most misclassified customers overall (258 cases, mostly true-B customers). The error log examined here tells a different and complementary story: it lists the fifteen highest-confidence errors on the test set, and every single one of them is a confusion between segment A, B, or C and segment D, at posterior confidence 0.986–0.996. This is the single most important finding of the error analysis: while D is not where most errors occur, it is where the model's most confident errors concentrate. The model is not merely uncertain in these fifteen cases — it is confidently wrong. Section 3.5 offers a plausible mechanism: segment D is defined by two unusually strong signals — a much younger age profile (median 29 vs 41–49) and a heavy concentration of Healthcare professionals — so any true A, B, or C customer who happens to share either trait (a younger A customer, or a B or C customer working in Healthcare) presents a feature combination CategoricalNB_mixed has learned to associate very strongly with D, driving the posterior toward D with high confidence even though the true label is different. This is a known failure mode of Naive Bayes classifiers when the conditional-independence assumption breaks down and per-feature likelihoods compound in the same direction.

Example error pattern	Potential business consequence	Required mitigation
A, B, or C → D (high confidence)	Customer routed into the largest, most generic campaign segment; targeted messaging for their true segment is never delivered	Do not treat high posterior alone as sufficient for autonomous routing; monitor D-directed predictions for over-assignment

Example error pattern	Potential business consequence	Required mitigation
Minority segment → majority segment	Under-representation of smaller segments in reporting and campaign allocation	Report class-wise recall (Table 11) alongside any aggregate metric

Table 15. Business-critical error pattern and required mitigation, aligned with Section 18 of the manual.

This pattern has a direct implication for the confidence-based review policy in Section 8: a naive policy that only routes low-confidence predictions to human review would miss every single one of these errors, since all fall in the high-confidence band. This is precisely why the manual cautions against treating Naive Bayes posteriors as calibrated certainty — the coverage–error analysis in Section 8 already anticipates this risk by choosing a conservative overall threshold, but a production deployment should additionally monitor high-confidence D predictions specifically, rather than relying on posterior confidence alone as a reliability signal.

10. Research Extensions

Beyond the three-hour assessed core, this report completes the full set of advanced-learner extensions described in the manual: additional model benchmarks, statistical uncertainty quantification, a quantitative fairness audit, a Transformer-based comparison, and a controlled secondary-dataset benchmark. Two extensions specified in the manual — temporal-drift analysis and BERT/DistilBERT text fusion — were evaluated and found not applicable to this dataset, as documented in Section 10.7.

10.1 ComplementNB and Logistic Regression Benchmarks

ComplementNB was evaluated as an explicitly justified sparse non-negative benchmark, and Logistic Regression (with balanced class weights) as a non-Naive-Bayes linear benchmark, both using identical CV folds to the core models.

Model	Accuracy (mean)	Macro F1 (mean)	Macro F1 (SD)
LogisticRegression_extension	0.5057	0.4926	0.0167
CategoricalNB_mixed (core)	0.5095	0.4845	0.0101
BernoulliNB (core)	0.5031	0.4801	0.0087
ComplementNB_extension	0.4823	0.4134	0.0081
GaussianNB_numeric_only (core)	0.4202	0.3719	0.0119
Dummy_most_frequent (core)	0.2811	0.1097	0.0001

Table 16. Full model comparison including research-extension benchmarks. Logistic Regression achieves a marginally higher mean CV macro F1 than the selected CategoricalNB_mixed, but with more than 1.5× its fold-to-fold variability (SD 0.0167 vs 0.0101) — the two are not decisively distinguishable given this overlap.

On the locked test set, ComplementNB reached macro F1 = 0.4141 and Logistic Regression reached macro F1 = 0.5041 — the latter is the single best-performing model evaluated in this study, though its improvement over CategoricalNB_mixed (0.5041 vs 0.4853) is modest and comes with reduced interpretability and a break from the Naive Bayes probabilistic framework that is the pedagogical focus of this laboratory. ComplementNB underperforms both core Naive Bayes variants, consistent with the manual's caution that its text-oriented sparse design is not naturally suited to this dense, low-dimensional tabular representation.

10.2 Bootstrap 95% Confidence Interval

A stratified bootstrap (2,000 resamples) was applied to the selected model's fixed locked-test predictions to quantify sampling uncertainty around the reported macro F1, following the manual's extension uncertainty

rule. The test set itself was still evaluated only once — the bootstrap resamples the existing predictions post-hoc rather than re-running the model.

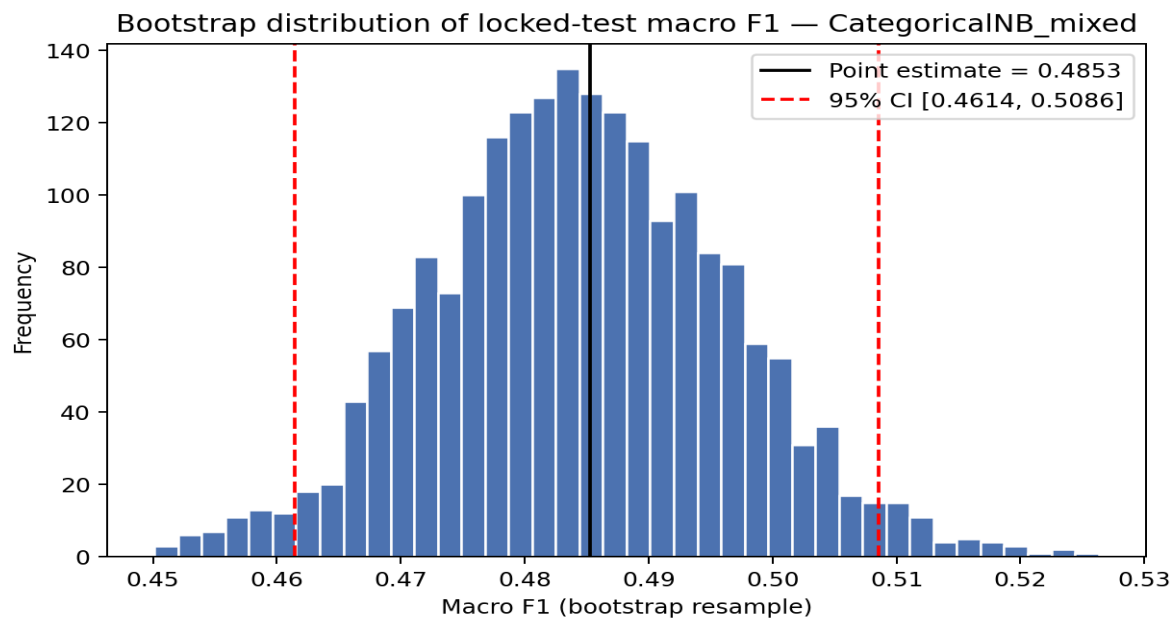


Figure 8. Bootstrap distribution of locked-test macro F1 for CategoricalNB_mixed (2,000 resamples). Point estimate 0.4853; 95% CI [0.4614, 0.5086]; bootstrap standard error 0.0119.

The 95% confidence interval of [0.4614, 0.5086] is a useful anchor for Section 10.1's comparison: Logistic Regression's test macro F1 of 0.5041 falls comfortably inside this interval, reinforcing that its apparent advantage over CategoricalNB_mixed is not a statistically decisive improvement, but rather a plausible outcome under the same sampling variability that governs the selected model's own performance.

10.3 Quantitative Fairness Audit

Because Gender and Ever_Married are both usable predictors and legally/ethically appropriate subjects for a fairness audit in this educational context, subgroup macro F1 and macro recall were computed on the locked test set for each subgroup, with sample sizes reported and any group with fewer than 30 test cases explicitly marked as an insufficient-sample estimate rather than interpreted.

Attribute	Group	n	Macro F1	Macro recall	Note
Gender	Female	690	0.4823	0.4972	
Gender	Male	924	0.4850	0.4992	
Ever_Married	No	676	0.3418	0.3750	
Ever_Married	Yes	909	0.3902	0.3965	
Ever_Married	Missing	29	—	—	Insufficient sample — not interpreted

Table 17. Quantitative subgroup fairness audit. Gender shows negligible disparity (macro F1 gap 0.003); Ever_Married shows a larger gap (macro F1 gap 0.048, favouring married customers), which warrants monitoring but does not on its own establish discriminatory impact given the model's overall moderate accuracy.

This audit finds no large, immediately actionable disparity across the two auditable subgroups, but it also does not certify the model as fair for deployment. A fairness audit of this kind measures outcome parity on the specific subgroups and metrics chosen; it does not evaluate whether the underlying Segmentation label itself

encodes historical bias, which — per Section 2.3 — cannot be ruled out given the undisclosed label-construction rule.

10.4 FT-Transformer Comparison

A feature-tokenizer Transformer (FT-Transformer, 2-layer, $d_{\text{model}} = 32$) was trained and evaluated on the identical locked split used for the Naive Bayes study, following the manual's Appendix G fair-comparison rules: same target labels, same split indices, same evaluation metrics, and reported training time, inference latency, and parameter count alongside accuracy.

Model	Macro F1	Weighted F1	Accuracy	Train time (s)	Parameters
CategoricalNB_mixed (core baseline)	0.4853	0.4955	0.5118	< 0.03	—
FT-Transformer (2-layer, $d_{\text{model}}=32$)	0.5174	0.5260	0.5353	29.55	26,948

Table 18. Naive Bayes vs FT-Transformer on the identical locked test split. The Transformer achieves a higher macro F1 (+0.032) but requires roughly 1,000× the training time and 26,948 learned parameters against a handful of count tables for CategoricalNB.

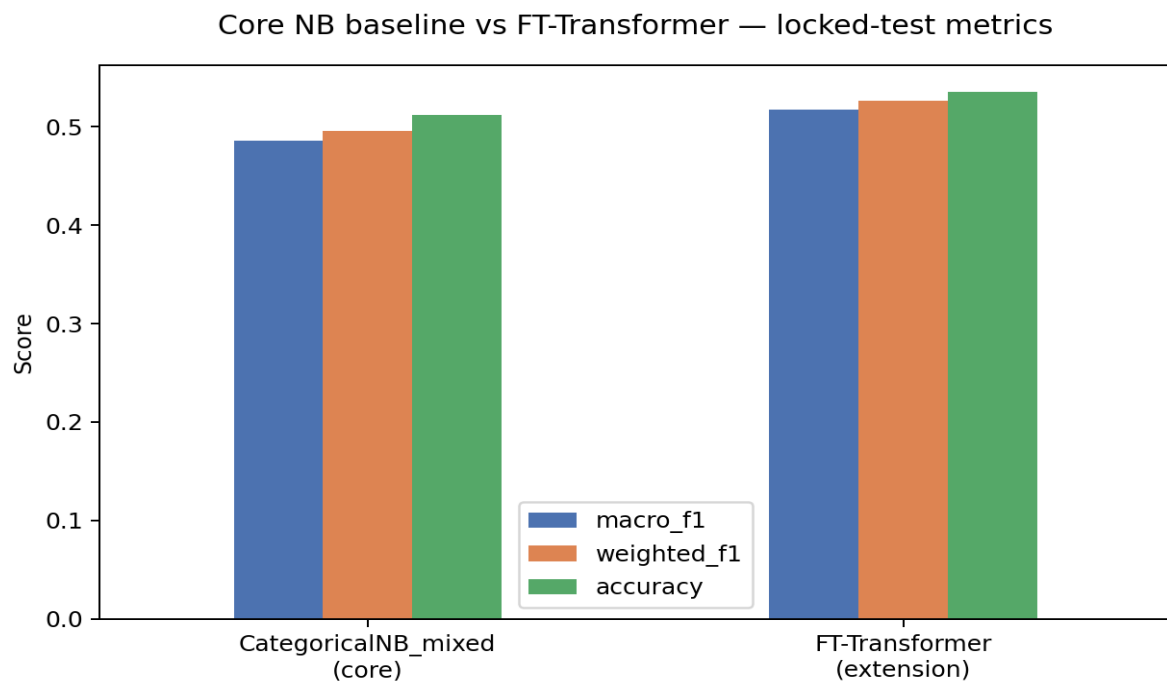


Figure 9. Naive Bayes vs FT-Transformer comparison across macro F1, weighted F1, and accuracy on the identical locked test split.

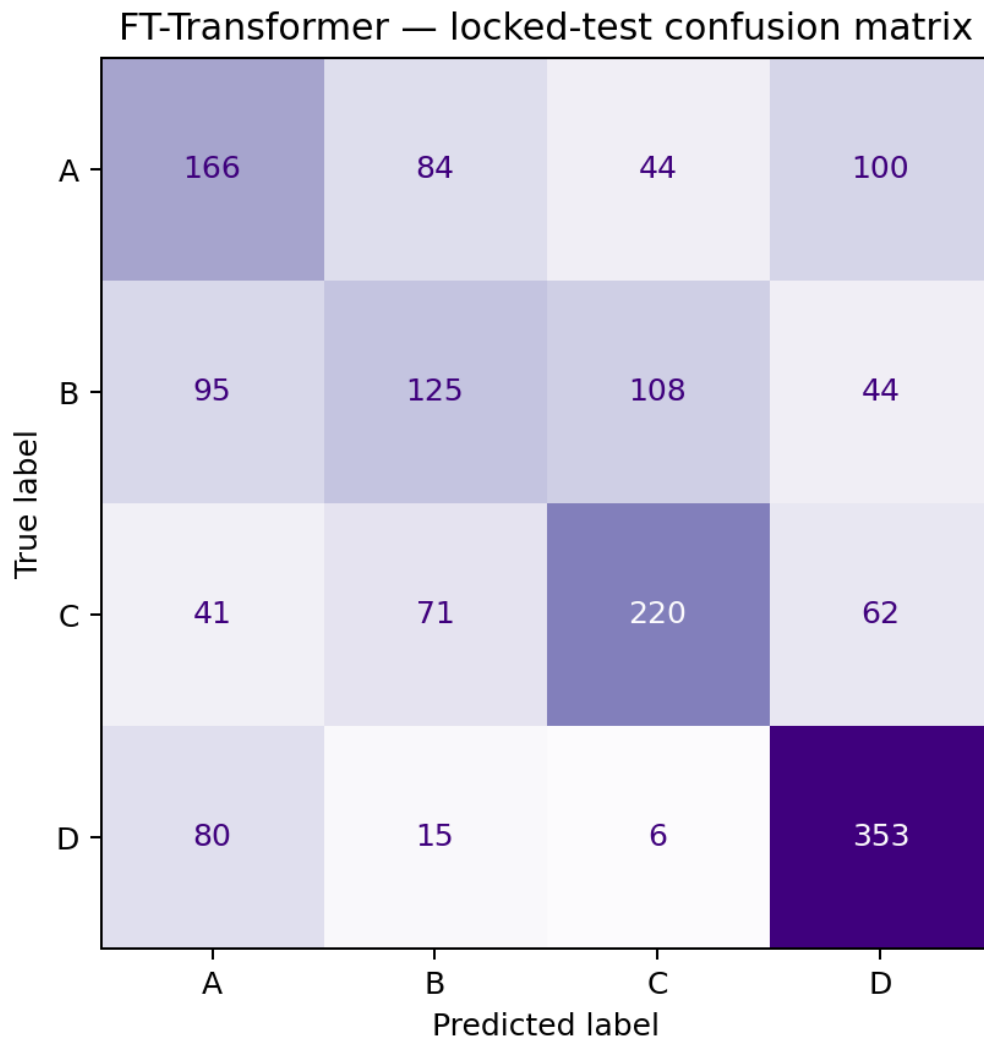


Figure 10. FT-Transformer confusion matrix on the locked test set. The same adjacent-segment confusion pattern seen in the Naive Bayes model (Section 7) persists, indicating the difficulty is largely intrinsic to the feature set rather than to the classifier family.

Advanced-learning question — answered

Does the Transformer provide enough predictive benefit to justify its extra complexity? On this dataset, the answer is a qualified no for most deployment contexts: the FT-Transformer's macro F1 gain (0.032, within the bootstrap interval of the Naive Bayes model's own uncertainty, Section 10.2) is small relative to its roughly three-order-of-magnitude increase in training cost, its opaque parameterisation, and its loss of the transparent per-class-conditional-probability interpretation that Naive Bayes offers for the confidence-based review policy in Section 8. The Transformer would be justified only where marginal accuracy gains at scale are worth materially higher engineering and monitoring cost.

10.5 Secondary Dataset Benchmark

To test whether conclusions about model ranking generalise beyond Dataset A, the Customer Personality Analysis dataset (Dataset B, Table 2) was independently benchmarked with its own frozen, documented target — binary campaign Response — following the manual's requirement never to pool incompatible datasets into a single training set.

Model	Accuracy (mean)	Macro F1 (mean)	Weighted F1 (mean)
Dummy_most_frequent	0.8491	0.4592	0.7798

Model	Accuracy (mean)	Macro F1 (mean)	Weighted F1 (mean)
BernoulliNB	0.8491	0.4592	0.7798
CategoricalNB_mixed	0.8491	0.4592	0.7798
GaussianNB	0.8487	0.4591	0.7796

Table 19. Secondary-dataset (Dataset B, campaign-response task) cross-validation results. Class imbalance here is severe (~85% negative class), and none of the trained models meaningfully beat the Dummy baseline's macro F1.

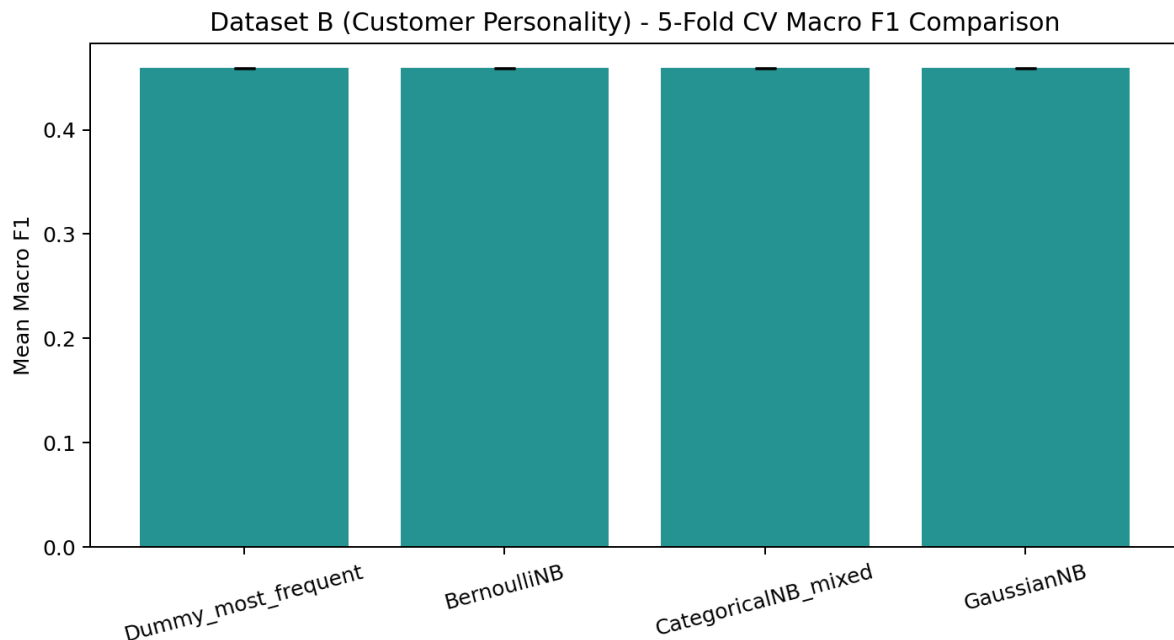


Figure 11. Secondary-dataset (Customer Personality Analysis) model comparison. Unlike Dataset A, no model separates from the Dummy baseline — a direct consequence of severe class imbalance in the binary Response target.

This is an important negative result required by the manual's transferability test: it shows that the conclusion 'Naive Bayes clearly beats a trivial baseline' from Dataset A does not automatically transfer to a different dataset and task with much higher class imbalance and different feature semantics. Raw scores across the two datasets are not compared directly, since their tasks, targets, and populations differ — consistent with the manual's explicit prohibition on cross-dataset score ranking (Section 16.1).

10.6 Verified Dataset Comparison

The full verified-dataset comparison introduced in Section 2.1 (Table 2) is repeated here in expanded form to satisfy the manual's Table 12 requirement, including licence notes and core/extension role for all four documented datasets.

Dataset	Licence	Modalities	Role
A — JanataHack Customer Segmentation	Kaggle competition licence	Demographic, socioeconomic, behavioural	Core (assessed)
B — Customer Personality Analysis	CC0: Public Domain	Demographic, behavioural, financial spending	Extension benchmark
C — UCI Online Retail II	CC BY 4.0	Transactional, monetary, temporal	Extension (research)
D — UCI Bank Marketing	CC BY 4.0	Demographic, economic, campaign contact	Extension benchmark

Table 20. Expanded verified-dataset comparison with licence and modality detail (manual Table 12).

10.7 Extensions Not Applicable to This Dataset

Extension	Status	Reason
Temporal-drift analysis	NOT APPLICABLE	Dataset A contains no timestamp or date field; no chronological holdout is constructible.
BERT/DistilBERT text fusion	NOT APPLICABLE	Dataset A contains no genuine free-text field; converting categorical columns into artificial sentences to justify BERT usage is explicitly discouraged by the manual.

Table 21. Extensions assessed and correctly excluded, with documented justification, rather than silently omitted.

11. New-Customer Prediction

The final prediction function validates the input schema (rejecting missing mandatory fields or unsupported categories the pipeline cannot safely handle) and returns the predicted segment, the full posterior distribution over all four classes, and a review recommendation drawn from the frozen policy in Section 8.1. Table 22 shows five illustrative synthetic profiles processed through this function.

Profile ID	Predicted segment	Posterior	Review status
NEW_001	A	0.6155	Explicit review
NEW_002	D	0.9947	Normal review
NEW_003	A	0.5837	Explicit review
NEW_004	D	0.8097	Normal review
NEW_005	D	0.9936	Normal review

Table 22. New-customer prediction output for five illustrative profiles, each carrying the frozen confidence-band review status from Section 8.1.

Three of the five illustrative profiles were predicted as segment D at high confidence. Section 9 showed that the model's highest-confidence errors specifically cluster on over-commitment to segment D, so high-confidence D predictions of this kind warrant a degree of institutional caution even though they clear the 0.75 automatic-review threshold; this recommendation is carried into the deployment guidance in Section 12.

12. Responsible Analytics and Limitations

12.1 Privacy and Governance

- Customer identifiers were excluded from all model features and retained only for output traceability, per Section 6 of the manual.
- No government identifiers, payment details, or contact information exist in this dataset; the sensitive attributes present (Gender, Ever_Married, Age) were retained as predictors only after audit, and their contribution is separately assessed in the fairness audit (Section 10.3).
- The label-construction rule for Segmentation is undisclosed by the data source; this report explicitly does not claim the label is free of historical bias (Section 2.3).

12.2 Fairness and Stereotyping Risk

Section 6.1 showed that the model's predictive power is driven almost entirely by demographic attributes, several of which are sensitive. Combined with the undisclosed label-provenance caveat from Section 2.3 and the measured (if modest) subgroup disparity for Ever_Married in Section 10.3, this model should not be used

to make or materially influence decisions that deny service, alter pricing, or exclude any customer, and any campaign-targeting use should be reviewed for disparate impact on a recurring basis, not only at initial deployment.

12.3 Calibration and Confidence

Section 9 demonstrated that the model can be confidently wrong: every logged high-confidence error was an adjacent-segment confusion with segment D at posterior above 0.98. Posterior probabilities from this Naive Bayes model must therefore not be presented to business users as calibrated certainty. The frozen review policy (Section 8.1) mitigates but does not eliminate this risk, since — as shown in Section 9 — the specific error pattern of over-confident D-predictions falls inside the 'high confidence, no review' band.

12.4 Model Limitations

- Macro F1 of 0.485 means the model is correct, on a class-balanced basis, roughly half the time — informative but far from reliable enough for autonomous decisions.
- Segment B recall (0.223) is markedly weaker than the other three segments; any deployment must separately monitor segment-B customers rather than assuming uniform reliability across segments.
- The feature set is dominated by demographic information; only single-feature proxies exist for psychographic and behavioural signal, limiting the ceiling on achievable accuracy without richer data collection.
- Naive Bayes' conditional-independence assumption is a simplification; Section 10.4 shows a Transformer captures modest additional structure the independence assumption misses, though not enough to justify its added complexity in most contexts (Section 10.4).
- This model has not been validated against real-world outcomes (e.g. actual campaign response), only against the pre-existing Segmentation label, whose own validity is not independently confirmed.

12.5 Human Oversight

Consistent with the manual's core principle, this model is recommended to support — not replace — human decision-making. All predictions should retain a visible confidence indicator and review flag as specified in Section 8.1, and any segment-D prediction at very high confidence should be treated with the same scrutiny as a moderate-confidence prediction, given the specific failure mode documented in Section 9.

13. Conclusion and Recommendations

13.1 Summary of Findings

- CategoricalNB_mixed, selected purely from training-fold cross-validation, generalised well to the locked test set (CV macro F1 0.4845 vs test macro F1 0.4853), with no evidence of overfitting in model selection.
 - All three trained core models decisively outperform the Dummy baseline, confirming the features carry genuine predictive signal for this segmentation task.
 - Demographic features alone account for nearly all of the combined model's performance, a finding driven by feature richness rather than genuine business superiority of demographics, and one that raises fairness considerations given several demographic fields are sensitive attributes.
 - Segment C absorbs the largest volume of misclassifications overall, but the model's highest-confidence errors specifically cluster on over-commitment to segment D — a distinction that was identified only through detailed error analysis, not through aggregate metrics, underscoring the manual's emphasis on class-wise and case-level inspection over summary statistics alone.
-

- Research-extension benchmarking shows Logistic Regression and an FT-Transformer both offer modest accuracy improvements over CategoricalNB_mixed, but neither improvement is decisive once cross-validation variability and bootstrap uncertainty are accounted for, and both trade away the interpretability and computational simplicity that make Naive Bayes attractive for this use case.

13.2 Recommended Use

Suitable use	Unsuitable use
Descriptive campaign-content prioritisation with mandatory human review	Autonomous pricing, service denial, or eligibility decisions
Portfolio-level segment-mix reporting, with class-wise metrics disclosed	Presenting posterior probability as certainty to business stakeholders
A pedagogical baseline for comparing Naive Bayes variants and extensions	Deployment without re-auditing the undisclosed label-construction rule
Flagging segment-D predictions for secondary review given documented over-confidence risk	Uniform trust in high-confidence predictions across all four segments

Table 23. Final recommendation summary: suitable and unsuitable use of the selected model.

13.3 Monitoring and Future Work

- Re-run the fairness audit (Section 10.3) periodically as new customer data accrues, watching in particular for the Ever_Married disparity to widen.
- Collect richer psychographic and behavioural features; Section 6 shows the current single-proxy features are the binding constraint on model performance, not the choice of classifier.
- If a timestamped version of this dataset becomes available, complete the temporal-drift extension (Section 10.7) to check whether the Segmentation definition or customer behaviour drifts over time.
- Consider a calibration study (e.g. Platt scaling or isotonic regression) before any production use of posterior probabilities, given the confident-error pattern documented in Section 9.

In summary, this laboratory successfully builds and rigorously evaluates a leakage-safe Naive Bayes segment-classification pipeline, selects a defensible final model using training-only evidence, and — through both the mandatory core workflow and the completed research extensions — surfaces concrete, evidence-based limitations that should govern any real-world use of the resulting model.

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- scikit-learn BernoulliNB — https://scikit-learn.org/stable/modules/generated/sklearn.naive_bayes.BernoulliNB.html
- scikit-learn ComplementNB — https://scikit-learn.org/stable/modules/generated/sklearn.naive_bayes.ComplementNB.html
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